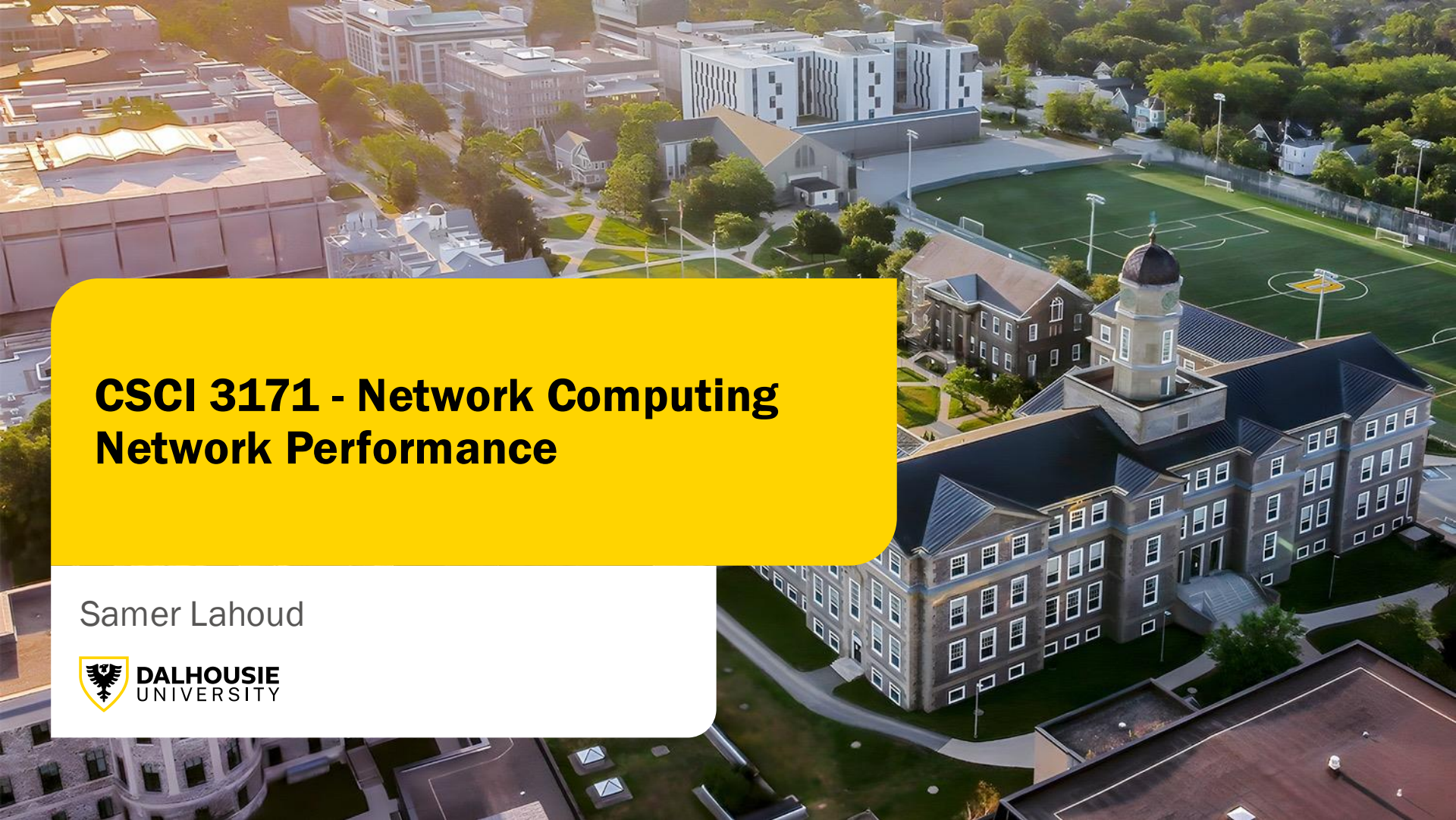


An aerial photograph of the Dalhousie University campus. In the foreground, a large, historic stone building with a central clock tower and many windows is visible. To its right is a green soccer field with white markings. In the background, there are more modern university buildings and lush green trees. A yellow banner is overlaid on the left side of the image, containing the course title.

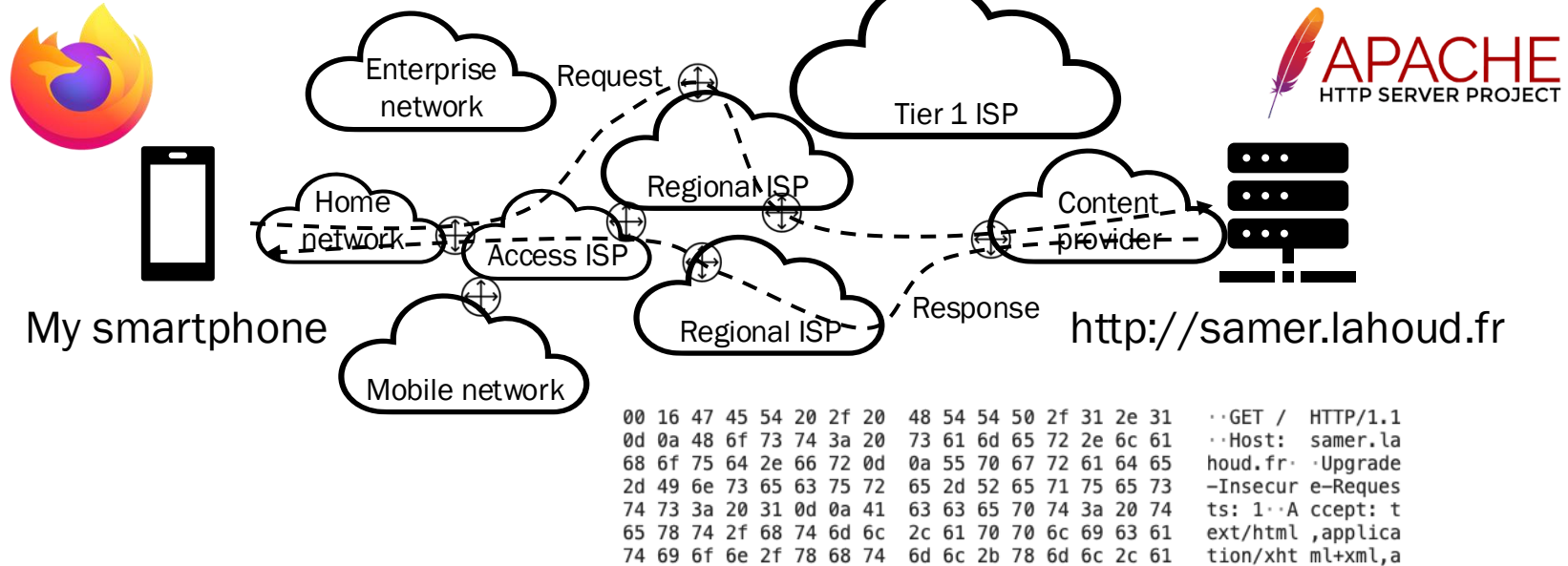
CSCI 3171 - Network Computing Network Performance

Samer Lahoud





Recap: Structure of the Internet



- Devices, services, applications, protocols, packet switches, networks, links



Goal and Roadmap

- Goal:
 - *Big picture* of the Internet
 - Introduction to terminology
- Overview and roadmap
 - What is the Internet? What is a protocol?
 - Network edge: hosts, access network, physical media
 - Network core: packet switching
 - Performance: loss, delay, throughput
 - Packet and circuit switching
 - Protocol layers, service models
 - Internet structure and challenges



Network Performance

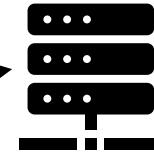


My smartphone

Request

```
00 16 47 45 54 20 2f 20 48 54 54 50 2f 31 2e 31  ..GET / HTTP/1.1
0d 0a 48 6f 73 74 3a 20 73 61 6d 65 72 2e 6c 61  ..Host: samer.la
68 6f 75 64 2e 66 72 0d 0a 55 70 67 72 61 64 65  houd.fr ·Upgrade
2d 49 6e 73 65 63 75 72 65 2d 52 65 71 75 65 73  -Insecur e-Request
74 73 3a 20 31 0d 0a 41 63 63 65 70 74 3a 20 74  ts: 1·A ccept: t
65 78 74 2f 68 74 6d 6c 2c 61 70 70 6c 69 63 61  ext/html ,applica
74 69 6f 6e 2f 78 68 74 6d 6c 2b 78 6d 6c 2c 61  tion/xht ml+xml,a
```

Response



 **APACHE**
HTTP SERVER PROJECT

<http://samer.lahoud.fr>

- We study the end-to-end *network* performance
 - How are applications messages sent over the Internet?
 - What are the performance indicators? How are they computed?



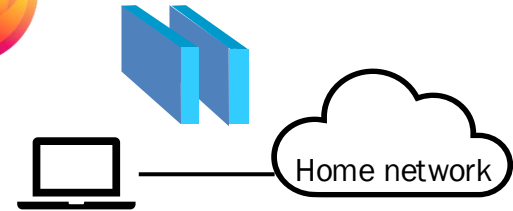
Host Sends Packets of Data

```
00 16 47 45 54 20 2f 20 48 54 54 50 2f 31 2e 31 GET / HTTP/1.1
0d 0a 48 6f 73 74 3a 20 73 61 6d 65 72 2e 6c 61 ..Host: samer.la
68 6f 75 64 2e 66 72 0d 0a 55 70 67 72 61 64 65 houd.fr Upgrade
2d 49 6e 73 65 63 75 72 65 2d 52 65 71 75 65 73 -Insecu e-Request
74 73 3a 20 31 0d 0a 41 63 63 65 70 74 3a 20 74 ts: 1..A ccept: t
65 78 74 2f 68 74 6d 6c 2c 61 70 70 6c 69 63 61 ext/html , applica
74 69 6f 6e 2f 78 68 74 6d 6c 2b 78 6d 6c 2c 61 tion/xht ml+xml,a
```

- Host sending functions
 - Takes application message
 - Breaks into smaller chunks, known as *packets*, of length L bits
 - Transmits packet into access network at transmission rate R
 - Link transmission rate, or link capacity, or link bandwidth



Two packets, L bits each



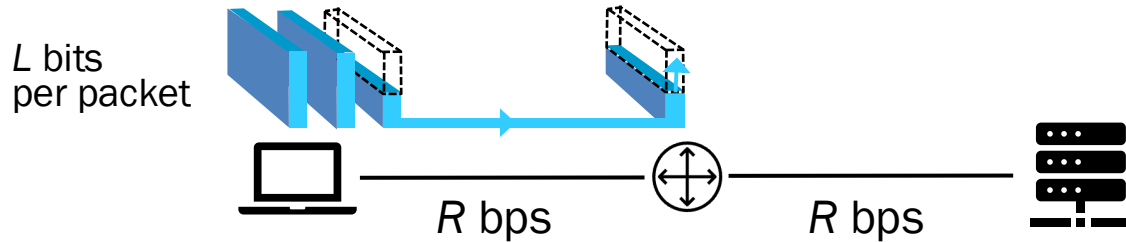
R : Link transmission rate

$$\text{packet transmission delay} = \frac{\text{time needed to transmit } L\text{-bit packet into link}}{L \text{ (bits)}} = \frac{L}{R} \text{ (bits/sec)}$$



Packet-Switching: Store-and-Forward

- Store-and-forward
 - Entire packet must arrive at router before it can be transmitted on next link
 - Two main benefits: error checking and transmission rate adaptation



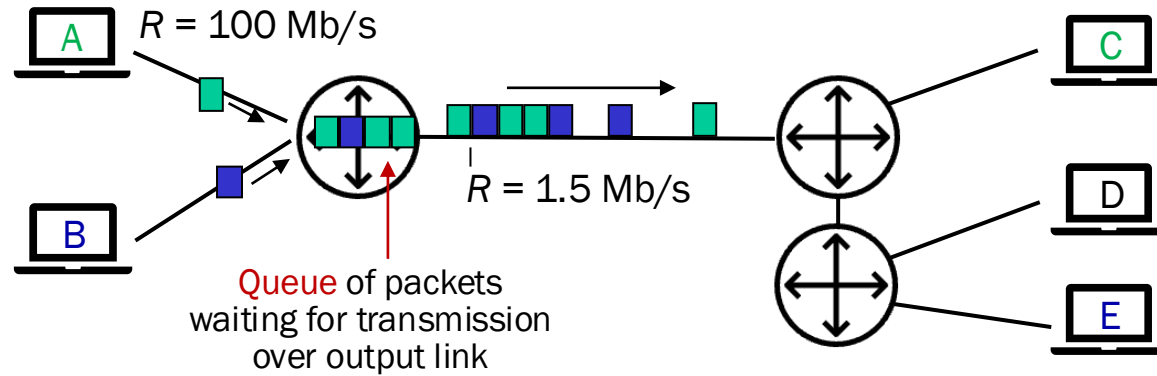
One-hop numerical example:

- $L = 10$ Kbits
- $R = 100$ Mbps
- One-hop transmission delay = 0.1 msec



Packet-Switching: Queueing

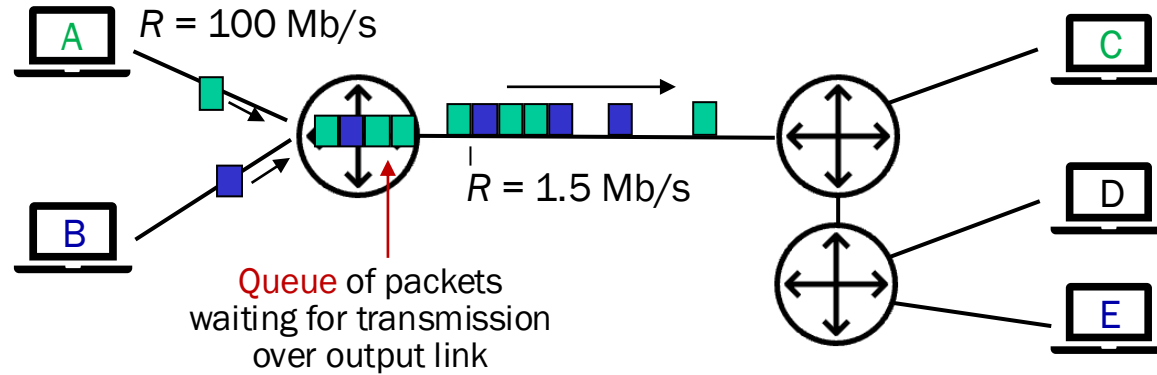
- Queueing occurs when work arrives faster than it can be serviced





Packet-Switching: Queueing

- Queueing occurs when work arrives faster than it can be serviced

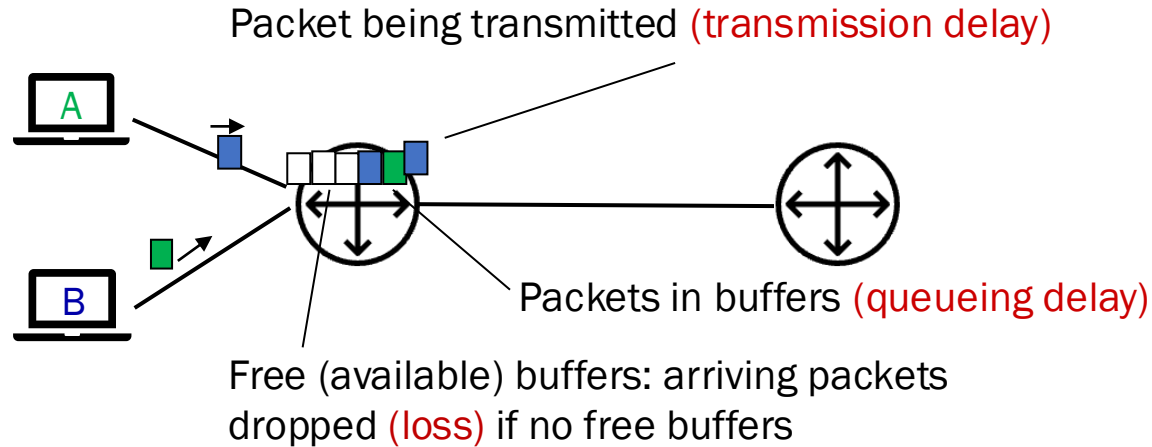


- What happens if arrival rate (in bps) to link exceeds transmission rate (bps) of link for some period of time?



How Do Packet Delay and Loss Occur?

- Packets queue in router buffers, waiting for turn for transmission
 - Queue length grows when arrival rate to link (temporarily) exceeds output link transmission rate
 - Packet loss occurs when memory to hold queued packets fills up



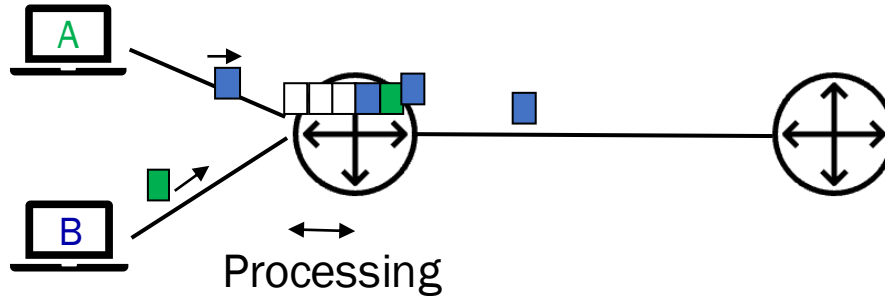


Network Performance Indicators

- Network delay
- Packet loss
- Throughput



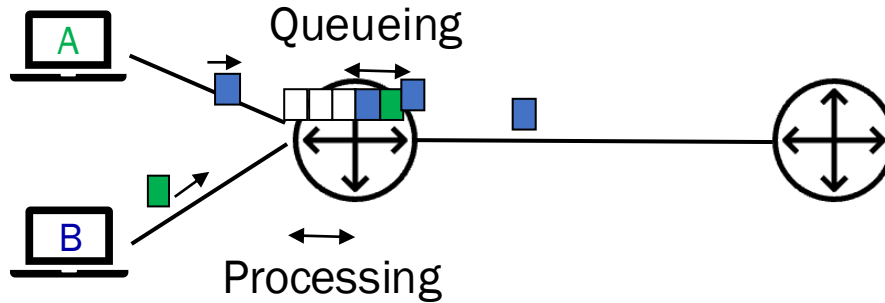
Packet Delay: Four Sources (1/4)



- Processing delay d_{proc} :
 - Check bit errors
 - Determine output link
 - Typically, less than 1 microsecond



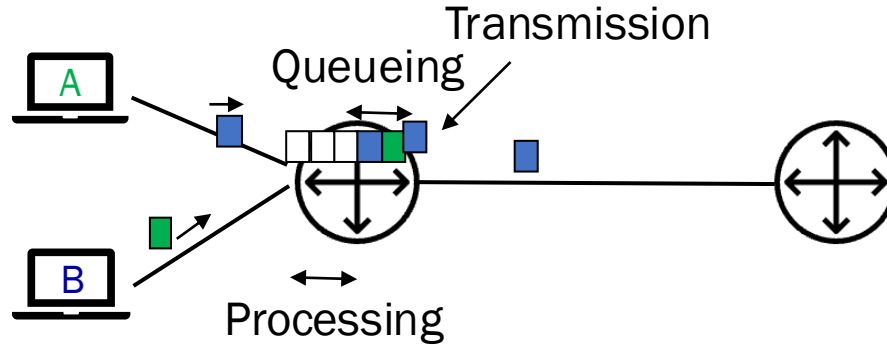
Packet Delay: Four Sources (2/4)



- Queueing delay d_{queue} :
 - Time waiting at output link for transmission
 - Depends on congestion level of router



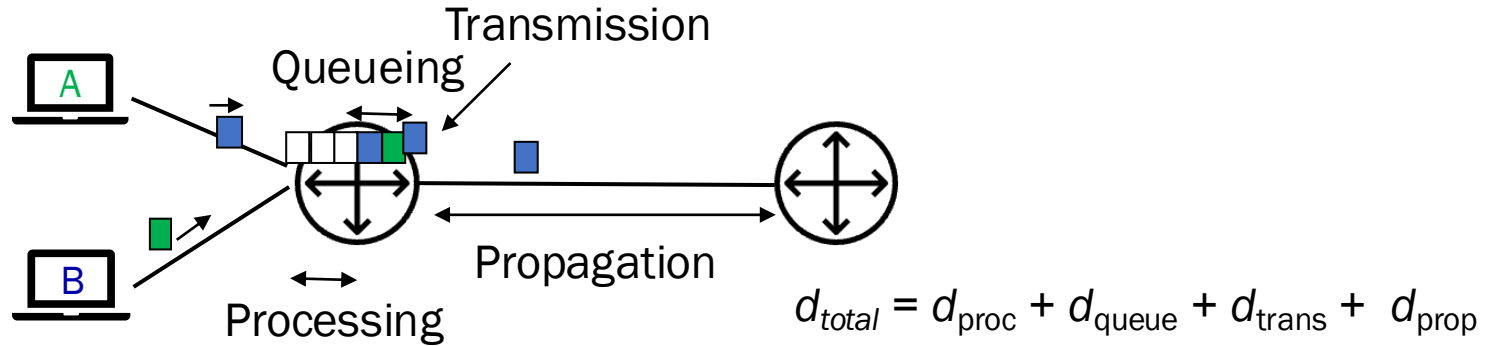
Packet Delay: Four Sources (3/4)



- Transmission delay d_{trans} :
 - L : packet length (bits)
 - R : link transmission rate (bps)
 - $d_{trans} = L/R$



Packet Delay: Four Sources (4/4)



- Propagation delay d_{prop} :
 - d : length of physical link
 - s : propagation speed ($\sim 2 \times 10^8$ m/sec)
 - $d_{prop} = d/s$



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Choose a slide to present

What is the Internet?

0 x

Processing delay

0 x

Queueing delay

0 ✓

Transmission delay

0 x

Propagation delay

→ Show correct answer





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SL 

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CSCI 3171 - W26



Choose a slide to present

What is the Internet?



0 

Processing delay

0 

Queueing delay

0 

Transmission delay

0 

Propagation delay

→ Show correct answer





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0 ✓

Processing delay

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Queueing delay

0 ✗

Transmission delay

0 ✗

Propagation delay

→ Show correct answer





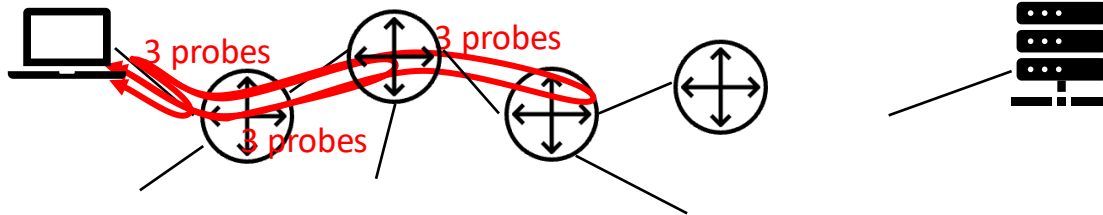
How to Reduce Delays in Networks?

- Processing
- Queueing
- Propagation
- Transmission



Delays and Routes on the Internet

- What do real Internet delay and loss look like?
- Traceroute program provides delay measurement from source to router along end-end Internet path towards destination.
- For all i :
 - Sends three packets that will reach router i on path towards destination
 - Router i will return packets to sender
 - Sender measures time interval between transmission and reply





Delays and Routes on the Internet

- Traceroute from timberlea.cs.dal.ca to www.epfl.edu

```
lahoud@timberlea:~$ traceroute www.epfl.edu
```

```
traceroute to www.epfl.edu (128.178.218.55), 30 hops max, 60 byte packets
```

```
1 GW81AD1600.Backbone.Dal.Ca (129.173.22.1) 0.371 ms 0.346 ms 0.332 ms
2 hlfx1rtr1.network.canarie.ca (205.189.32.233) 0.943 ms 1.227 ms 1.491 ms
3 mtr13rtr1.network.canarie.ca (205.189.32.48) 15.355 ms 15.343 ms 15.070 ms
4 mtr12rtr1.network.canarie.ca (205.189.32.200) 18.579 ms 36.517 ms 36.671 ms
5 canarie-bckp.mx1.par.fr.geant.net (62.40.124.225) 100.589 ms 100.564 ms 100.120 ms
6 ae7.mx1.gen.ch.geant.net (62.40.98.238) 108.329 ms 108.318 ms 108.540 ms
7 swice1-100ge-0-3-0-1.switch.ch (62.40.124.22) 108.908 ms 108.704 ms 108.886 ms
8 swiGE3-100GE-0-0-1-0.switch.ch (130.59.38.194) 108.588 ms 108.649 ms 108.775 ms
9 swiEL3-100GE-0-0-0-2.switch.ch (130.59.36.81) 109.715 ms 109.484 ms 109.364 ms
10 cv-internet-swiel3-18.epfl.ch (192.33.209.18) 108.965 ms 109.189 ms 109.168 ms
```

3 measurement
delays from timberlea
to hlfx1rtr1

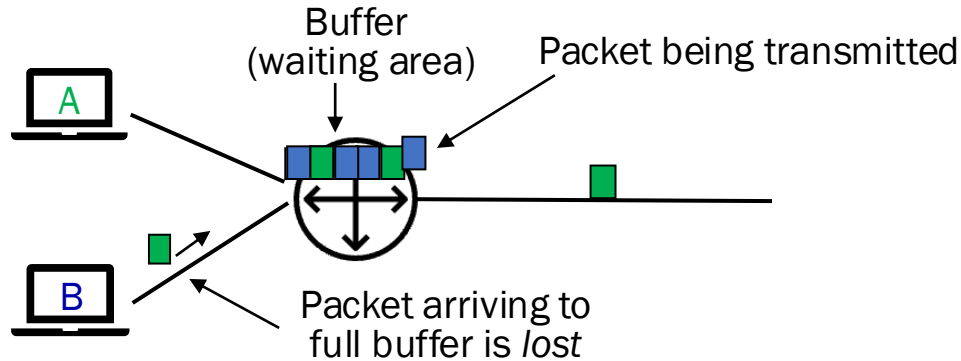
3

Trans-oceanic link



Packet Loss

- Queue (aka buffer) has finite capacity
- Packet arriving to full queue dropped (lost)
 - Lost packet may be retransmitted by previous node, by source end system, or not at all



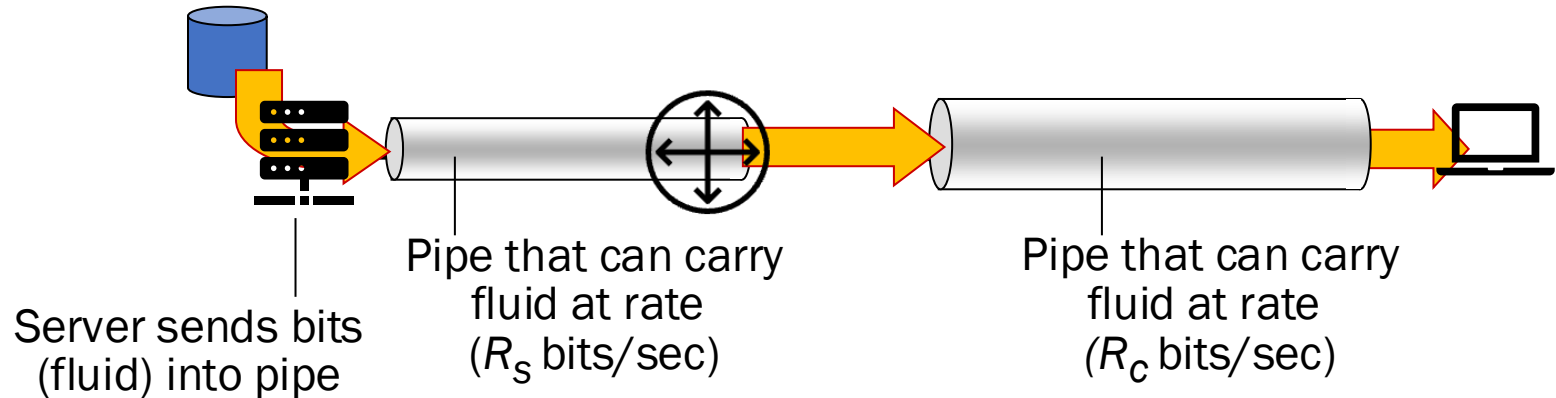
Check out the Java applet for an interactive animation

<https://computerscience.unicam.it/marcantoni/reti/applet/QueuingAndLossInteractive/1.html>



Throughput

- Throughput: rate (bits/time unit) at which bits are being sent from sender to receiver
 - Instantaneous: rate at given point in time
 - Average: rate over longer period of time

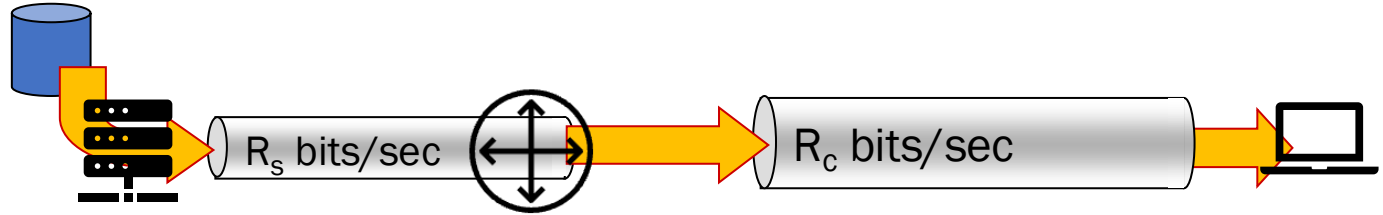




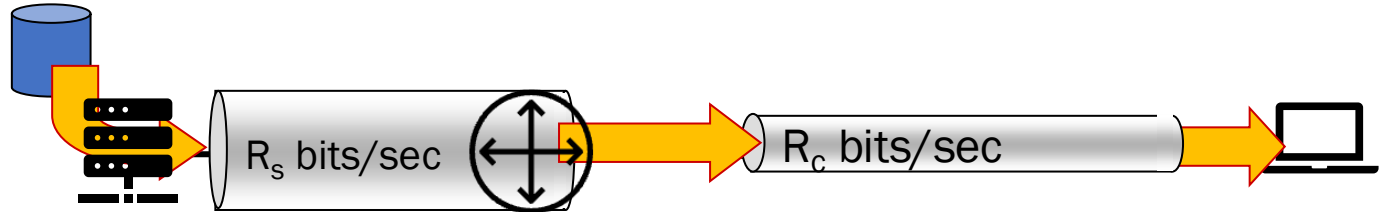
Throughput and Bottleneck

- Bottleneck link: link on end-end path that constrains end-end throughput
- What is the average end-to-end throughput?

- $R_s < R_c$



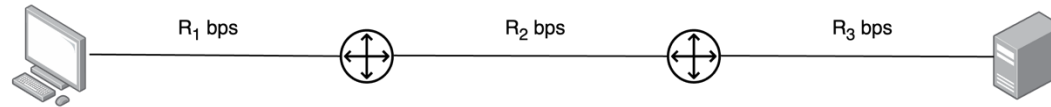
- $R_s > R_c$





Performance Computation Example

- Three-link example
- Transmission rates: R_i (bps)
- Distances: d_i (m)
- Packet size: P (bits)





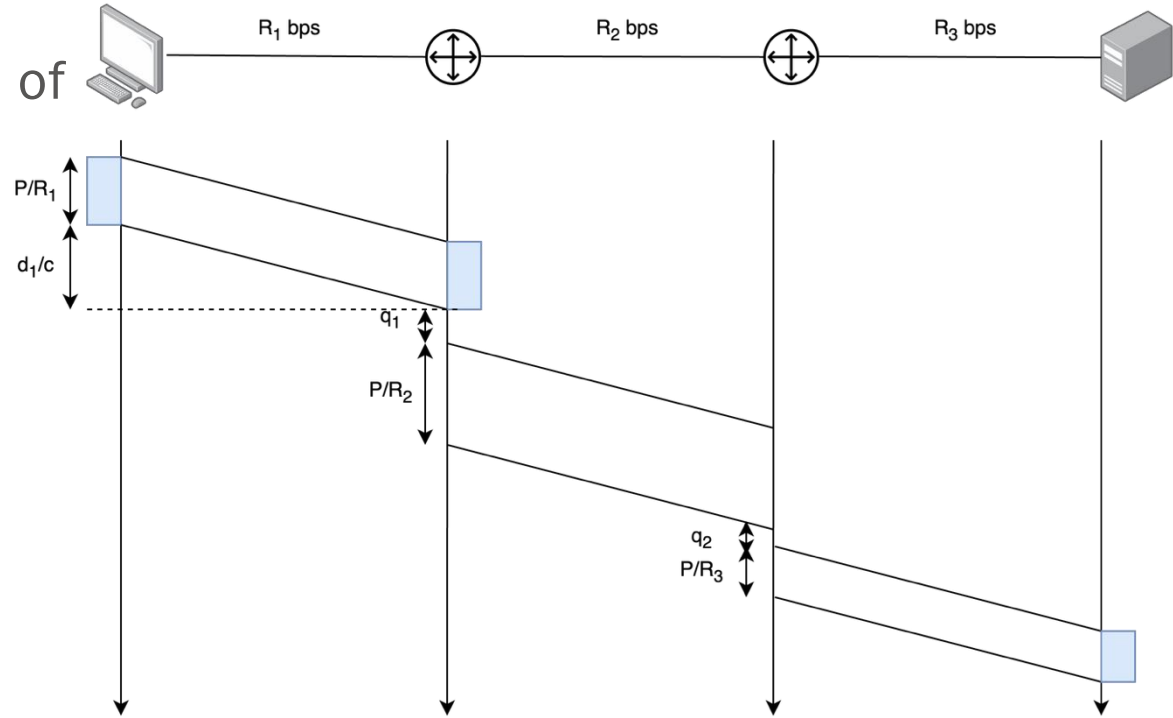
End-to-End Delay

- End-to-end delay = sum of

- Transmission delays
 - P / R_i for $i=1,2,3$
- Propagation delays
 - d_i / c for $i=1,2,3$
- Processing delays
- Queueing delays

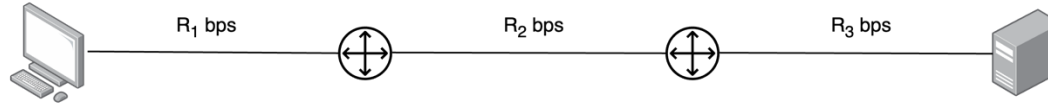
- Observations:

- Store and forward mechanism
- $R_2 < R_1$





Bottleneck and Throughput

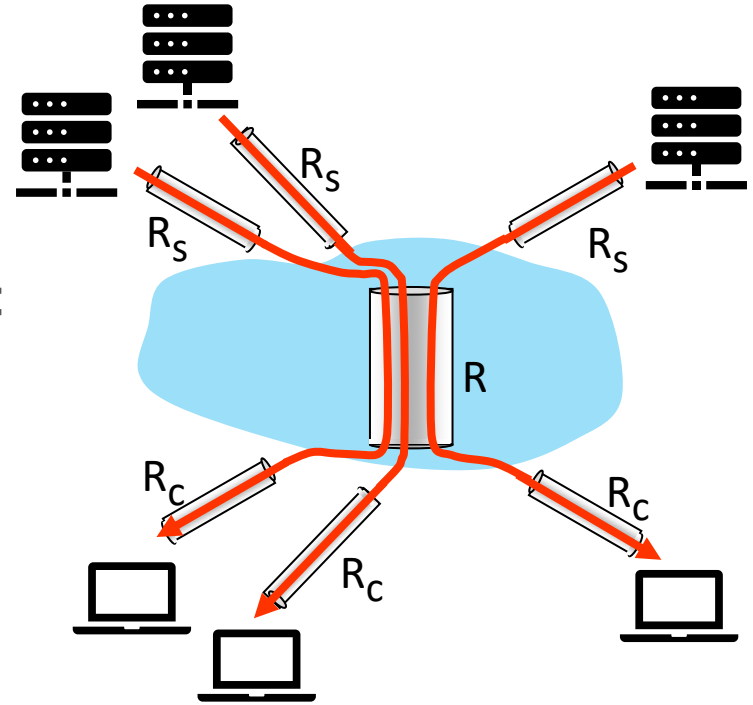


- Assuming $R_2 < R_1$ and $R_2 < R_3$,
 - Link 2 is the bottleneck
 - End-to-end throughput is R_2 bps
- Utilization ratio of link 1 is equal to 1
 - Packets are waiting in the queue on the first router
- Utilization ratio of link 2 is equal to 1
- Utilization ratio of link 3 is equal to R_2/R_3



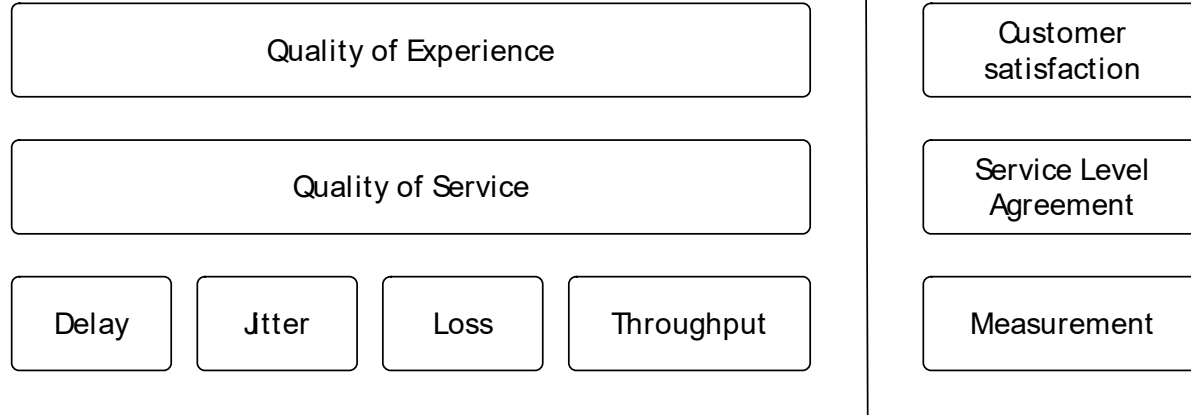
Throughput: Network Scenario

- Where is the bottleneck?
 - Minimum of R_c , R_s and $R/3$
- End-end throughput of one connection:
 - $\min(R_c, R_s, R/3)$



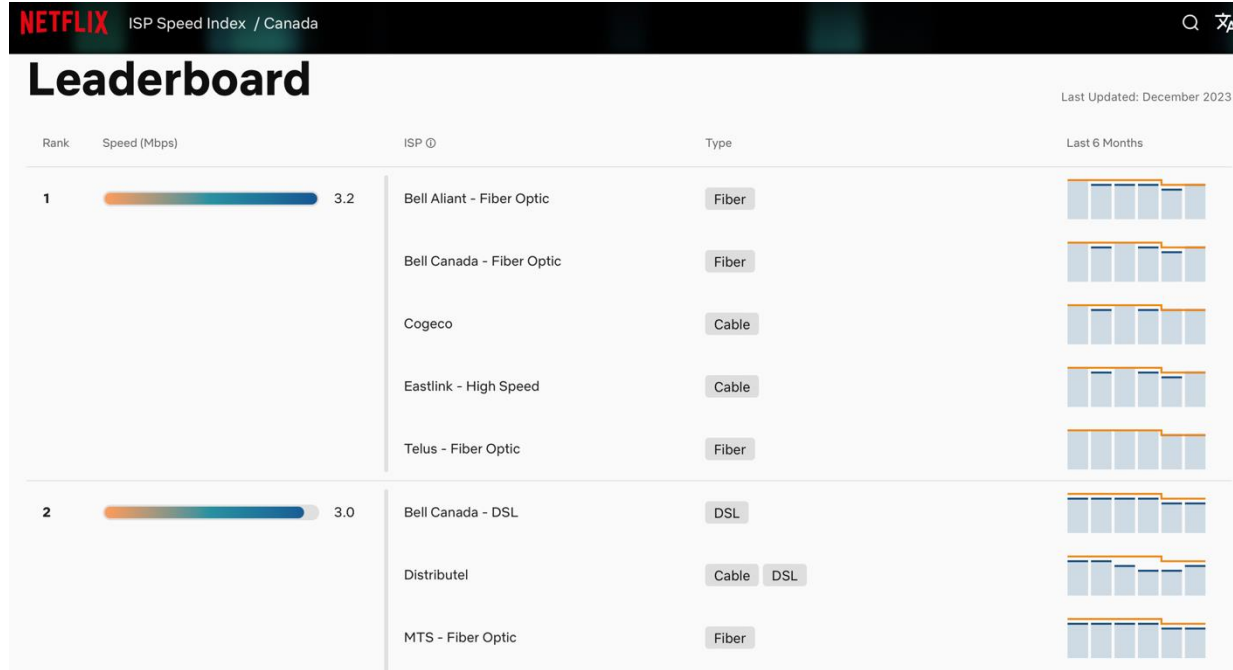


Performance and Quality





Performance is a Service Differentiator





Applications and Requirements

Application	Packet loss tolerance	Throughput requirement	Delay tolerance
File transfer (including email, and web browsing)	No	Elastic	Yes
Voice chat	Yes	[Few kbps, 1 Mbps]	100 msec
Video streaming	Yes	[Few kbps, 5 Mbps]	Few seconds
Gaming	Yes	Few kbps	100 msec